Accelerating Innovative Energy Solutions Using Combustion Simulations

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Combustion-based transportation, electricity generation, and industrial heating in manufacturing constitute the three largest sectors of energy demand. Some of the recent technology development in these sectors are: switching to alternative fuels for the transportation sector, increasing energy efficiency in the power sector, and capturing carbon emissions from conventional power generators. Several teams at the National Renewable Energy Laboratory have been actively advancing research in these areas by leveraging computational modeling of combustion processes across the heavy-duty land-based transportation, aviation, and power generation sectors. This article summarizes some of these efforts, demonstrating the potential of advanced computational techniques to generate technological solutions that will transform the global energy system.

The *scale* and *urgency* of optimizing the global energy system, including electricity generation, transportation, and industrial processes requires a critical reexamination of conventional technologies. Meeting the increased energy demand from artificial intelligence (AI) data centers will require massive scale up in deployment of natural gas, nuclear, wind, solar, hydro, and other energy technologies. Other technology changes that will make a significant impact, such as switching to low-carbon fuels or increasing the energy efficiency of existing systems, require significant re-engineering of systems that have been optimized over decades of use. Additionally, enabling carbon capture, utilization, and storage (CCUS) will require large changes to combustion systems. Developing these revolutionary new technologies purely through experimental design exploration is no longer a viable approach, both in terms of costs and time required to mature these technologies. Advanced simulation is now seen as a critical tool for developing and deploying new technologies such as energy storage and energy-efficient manufacturing processes. These simulations often leverage machine learning (ML) and AI to further accelerate technology development.[1] In a change from the existing paradigm, new industrial combustion systems that either reduce emissions in sectors such as transportation, enable adoption of varied fuels, or even allow carbon-negative combustion are being *enabled* and *optimized* using advanced simulation, particularly numerical simulations of reacting flow.

Fortunately, the computing hardware and software required to meet this need are becoming available simultaneously. In the past ten years, there has been a 30-fold increase in the computational resources available for U.S. Department of Energy (DOE)-funded energy research.[1] Exascale computing facilities, taking advantage of GPU accelerators, are now being routinely exploited for fundamental research. As GPU-enabled architectures become the standard in high-performance computing, and cascade into workstation computing, the use of high-fidelity simulation as a tool for understanding basic physics of reacting flow systems, exploring corresponding design spaces, and de-risking new technologies is expected to increase. In the following section, we present several exemplary application areas under development at

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National Renewable Energy Laboratory (NREL) where advanced combustion simulations are critical to ongoing maturation of novel technologies targeting optimization of key energy systems.

HIGH-FIDELITY SIMULATIONS OF COMBUSTION

The DOE's Exascale Computing Program (ECP) featured a codevelopment approach to enable a wide range of application domains to utilize GPU-based computing hardware. The project supported simultaneous and interleaved development of compilers, programming models, data structures, and solver libraries for exascale hardware, together with 24 domain applications of specific interest to the DOE. Under this program, NREL, in partnership with Sandia National Laboratories and Oak Ridge National Laboratory, developed the Pele software suite[2] for turbulent combustion and other thermochemically complex flows. The Pele suite features compressible[3] and low Mach[4] reacting flow solvers, which leverage a common library of models for thermodynamics, transport, spray, soot, thermal radiation, and finite-rate chemistry. For reacting flow simulations, compressible flow solvers resolve local density variations resulting from spatially varying pressure and temperature fields. Conversely, low-Mach number solvers account for density variations as a function of spatially local temperature field and a globally defined thermodynamic pressure. The suite leverages the AMReX library to enable performance portable adaptive mesh refinement, Lagrangian particle tracking, and flexible memory and data structure management.

The Pele suite of codes have benefited from many advances in numerical methods and the use of performance portability libraries, codeveloped through ECP. The use of the performance portability abstractions in AMReX enable simulation performance across all major GPU vendors. We performed and documented extensive tuning campaigns to establish best practices for the work unit size (e.g., thread counts for kernel launches) across a variety of architectures. These campaigns also included deep investigations of linear solver performance (for the incompressible code) and ordinary differential equation solvers (for the chemistry, leveraging SUNDIALS). We established best practices for the combustion systems of interest in terms of, for example, types of solvers, smoothers, and preconditioners. Numerical algorithm work included the research and development of new discretization techniques for complex geometry in the context of adaptive mesh refinement. A cornerstone enabling and sustaining this work has been the development and adherence to software engineering best practices: version control, third party libraries indexed as submodules, automated testing suites through continuous integration on cloud servers to check for performance and simulation regressions, strong linting checks to enforce up-to-date code development best practices, and automated documentation. The reader is referred to the relevant publications for more details.[2,3,4,5,6]

Application targets for the Pele suite include a wide range of energy generation technologies, such as the deployment of sustainable aviation turbine fuel (SATF) through assessing the impact of fuel properties of SATF in realistic combustor geometries, replacing hydrocarbon combustion technologies in industrial processes with hydrogen combustion or hydrogen thermal plasma arcs, and de-risking alternative combustion technologies such as the Allam–Fetvedt cycle. In these examples, the exascale-enabled Pele suite will deliver previously intractable simulations, transforming simulations previously considered to be prohibitively computationally expensive into routine evaluations of high-fidelity models. In addition to speeding up individual simulations, increases in available computing resources are enabling massive ensemble computations required to develop on-the-fly, auto-validated reduced order models, that are critical to many outer-loop decision making applications. For combustion applications, the increased fidelity of model predictions enables a more complete accounting of intricate chemical processes and the associated impact of turbulent fluid dynamics to be incorporated into high-dimensional parameter scans, advanced control strategies, uncertainty quantification, and inverse design, ultimately accelerating the design and optimization of advanced reacting flow technologies.

HEAVY-DUTY TRANSPORTATION: CO-OPTIMA

In 2016 the DOE Bioenergy Technologies Office and Vehicle Technologies Office launched an initiative to accelerate the introduction of affordable, scalable, and synthetic high-performance fuels for use in high-efficiency, low-emission engines.[a] This program, known as Co-Optimization of Fuels and Engines (Co-Optima), was focused on reducing the carbon footprint of land-based transportation sectors. Researchers from nine national labs and 40 partnering institutions investigated bio-blendstocks to be blended into gasoline and diesel—in conjunction with advanced combustion

[a]https://www.energy.gov/eere/bioenergy/co-optimization-fuels-engines

techniques to optimize fuel utilization and reduce emissions. Discovering these strategies using experimental methods requires large amounts of fuels and numerous trials with engine designs leading to prohibitive rise in experimental costs. Furthermore, conducting such a study would require infrastructure and facilities only available at a small number of institutions, which dramatically limits the ability to parallelize exploration of novel solutions. High-fidelity simulations coupled with innovative ML techniques provide a novel approach to co-optimize fuel and engine design. As discussed next, NREL researchers have demonstrated the use of reinforcement learning (RL) techniques to discover new system operational strategies to optimize the combustion of proposed diesel fuels.

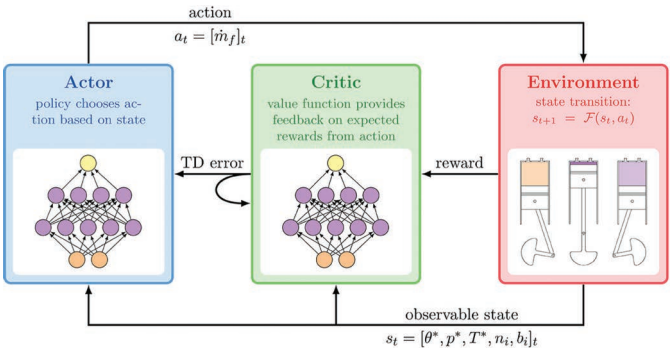

FIGURE 1. Actor–critic RL architecture, with included specifics of the engine framework explored in this work. The actor implements a policy, which provides actions to the environment. The critic implements the value function and provides an error for the actor. The environment provides the states to the actor and the critic and the reward to the critic. Figure 1 from Henry de Frahan et al.,[7] reproduced with permission.

Reinforcement Learning-Based Co-Optimization of Engine Design and Fuel Chemistry

A key goal of the Co-Optima program was to develop new methods for joint optimization of the performance of hard-to-electrify engines and various combinations of fuels used within. Specific focus targeted heavy-duty and off-road transportation vehicles, which predominantly rely upon compression-ignition (CI) engines. In CI engines fuel is injected directly into a combustion cylinder and compressive heating, induced by a rotating piston, helps to auto-ignite the fuel-air mixture. The highly complex auto-ignition sequence is extremely sensitive to engine design features, operating conditions, and the unique fuel blends injected into the cylinder. In recent years, more advanced CI strategies have been developed that use a series of discrete fuel injections and multiple fuels and fuel blends to control the combustion phasing and reduce harmful emissions. These new ignition methods require a concomitant redesign of engine parameters (such as compression ratios) that are tailored to specific novel fuel blends, including various biofuels, and the corresponding design space for optimization has grown drastically. For these systems, traditional optimization techniques are stretched beyond their limits. Under the Co-Optima project, NREL developed and deployed cutting-edge techniques based on ML and AI. Specifically, RL is a method that is well-suited for combinatorial optimization problems with high-dimensional parameter spaces to explore. The Co-Optima team implemented a RL agent to interact with a simulation environment of a CI engine.

The general process of the RL algorithm is shown in Figure 1, and can be summarized in a seven-step process: 1) the RL agent is initialized with a pseudorandom injection strategy policy, 2) the RL agent receives input information about the current state of the engine environment (e.g., temperature, pressure, crank angle, and so on), and 3) chooses to inject a fuel (from a list of candidate fuels). That action is passed along to the engine environment, which then 4) performs the chosen action and evaluates the dynamics of the engine for one time evolution step. 5) A process "critic" determines the reward (e.g., penalized emissions and rewarded engine work) to be associated with the action that was taken, and 6) the next engine state and reward from the previous action are passed to the agent. 7) steps 2–6 repeat and, periodically, the agent policy function is updated, which allows the RL agent to "learn" the optimal control policy.

During the Co-Optima project, the NREL team demonstrated the power of the RL strategy in two studies. The first[7] showed that an RL agent could learn a dynamic control strategy for a multipulse CI engine, based on a hierarchy of models with increasing fidelities for the physics and chemical reaction dynamics. The RL agent was also deployed for the engine environment with a dynamic reward function that evaluates the tradeoff between engine work output and reduction in out-of-cylinder emissions. The RL agent was shown able to dynamically alternate between emissions reduction and maximized work output, based on the (dynamic) needs of the application. These results proved the power of the method and showcased the promise for optimization in many complex control environments.

The second study[8] expanded the analysis to multipulse and multifuel operating conditions. The team

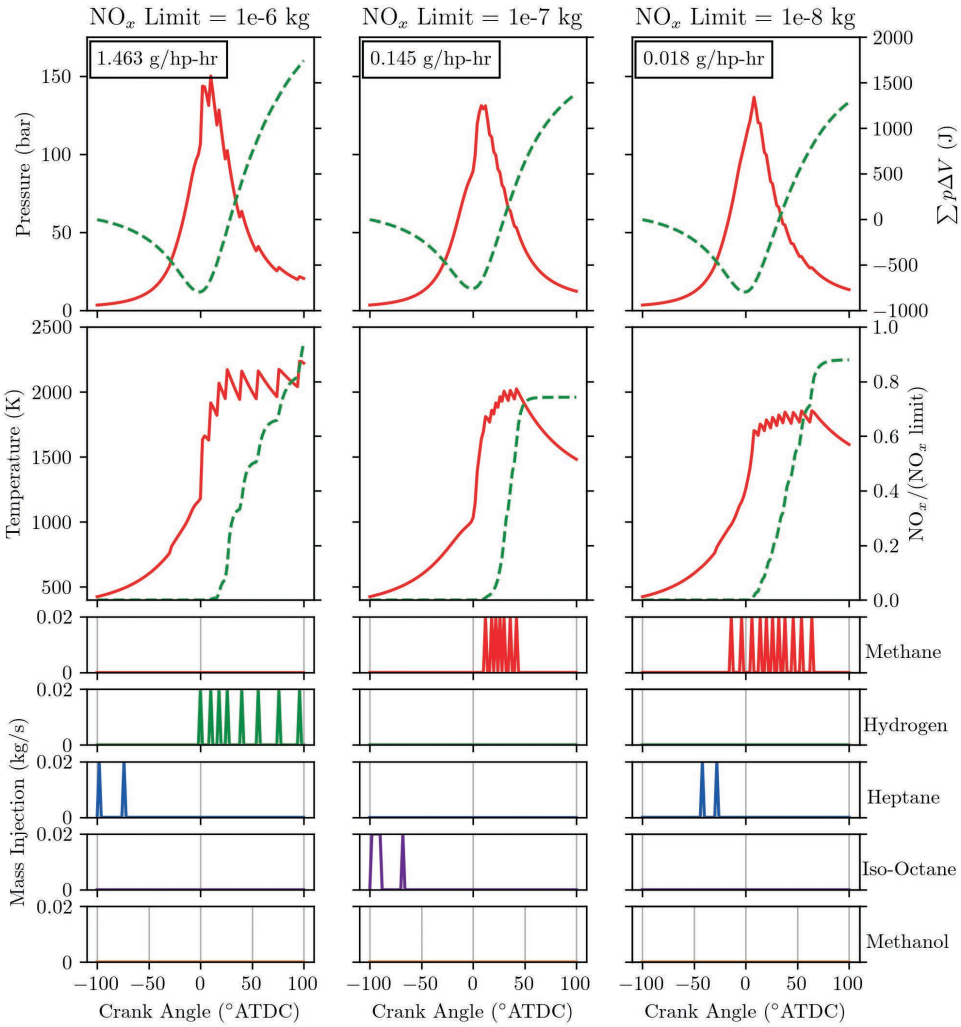

FIGURE 2. Evaluation of the RL agent on three different NOx emission criteria, demonstrating the ability of the RL method to discover distinct multifuel strategies under changing emissions targets. Figure 1 from Wimer et al.,[8] reproduced with permission.

allowed the RL agent to select from a list of five fuels to inject throughout the engine cycle with the goal of the optimization to maximize the amount of engine work produced while staying below a series of ever-increasing NOx emissions restrictions that were loosely based on existing regulations at the federal and state level. By evaluating the RL optimal control strategy under a series of emissions targets, the team was able to proactively examine the fuel characteristics that would be required for future scenarios.

Figure 2 shows the results of the RL agent's optimization after the end of training for three different NOx emissions regulation targets. The bottom half of the figure shows the final injection schedule for each of the five candidate fuels inspected. When the NOx emissions targets are forgiving, the optimal injection strategy is to preinject heptane followed by near top-dead-center (TDC) injections of hydrogen. As the NOx emissions target becomes more strict, the RL agent adapts automatically to a dual-fuel mixture of iso-octane/heptane and methane.

The studies performed under the AI/ML/RL Co-Optima thrust indicate that these RL-based methods can be readily applied to a wide range of engineering applications that involve complex optimization and control challenges. Specifically, the approach can be used for SATF studies where a wide range of candidate fuels need to be screened as drop-in replacements for jet fuel. The method can also be deployed as part of a transformation of global industrial heating effort, where the ability to optimize a specific application is its own multidimensional problem, complete with

complex physics, numerous control parameters, and a dynamic optimization function that likely depends upon process efficiency and dynamically shifting emissions reductions targets.

One of the primary benefits from AI/ML techniques in simulation and modeling over conventional strategies is the characteristic that these methods tend to perform well when extrapolated to unseen conditions. Once trained upon a subset of the parameter space, the RL agent can often be applied to unseen regions and exhibit acceptable accuracy.[8] This characteristic is particularly appealing when considering the shifting landscape of industrial heating, where strategies are directly linked to emissions regulation targets that can shift and require frequent redesigns of expensive engineering systems. AI/ML methods can be readily trained on a subset of the parameter space and then used to evaluate the impacts of evolving constraints (e.g., changing emissions targets).

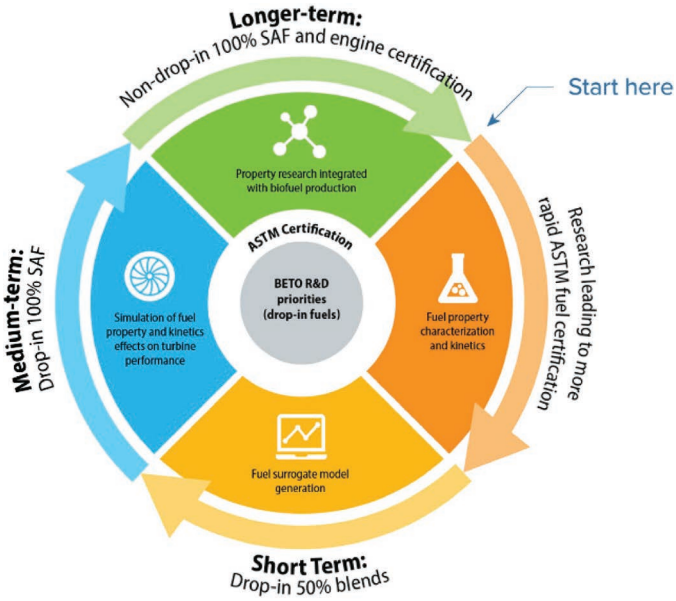

FIGURE 3. SATF technology research landscape at NREL.

SUSTAINABLE AVIATION TURBINE FUELS

Commercial aviation is one of the hardest sectors to introduce new technologies due to strict regulation, infrastructure issues, and decades-long design cycle times. Liquid hydrocarbon fuels remain the only viable energy carrier for commercial aviation, particularly for long distances, high capacities, and higher-speed flight, and are likely to remain so for many years. To fully realize the potential of SATF, a federal-government-wide grand challenge has been established to expand the SATF production, increasing production capabilities from current annual levels of 4.5 million gallons by nearly three orders of magnitude, to 3 billion gallons.[b] Currently, there are eight fuels produced from various sustainable feedstocks that have been certified for use in commercial aviation. However, all of these so-called "drop-in fuels" must be blended with a minimum of 50% petroleum-based jet fuel. To meet the SATF grand challenge, drop-in fuels will need to be certified at 100% (i.e., with no petroleum fuel blending). They will also need corresponding production processes to be scaled up to millions of gallons. Additionally, there is a huge market demand from the end-use airlines to burn 100% drop-in SATF. In response, American Society for Testing and Materials (ASTM) has commissioned a task force for approval of 100% SATF. Unfortunately, the number of new fuels needed for certification presents a major financial and technical challenge. High-fidelity simulations can play a significant role in helping to de-risk this certification; simulation can provide an alternative to laboratory-based screening fuel candidates that is both time- and cost-effective.

Figure 3 shows a schematic of an NREL research ecosystem recently established to de-risk certification new SATF fuels at the required higher blend ratios. Within this cross-lab collaborative research program, any new fuel under development is fully characterized, and its performance evaluated for representative aircraft engine combustors using the Pele high-fidelity predictive simulations. The performance of the fuels is communicated to the bio-fuel production teams who can tune the fuel composition further to satisfy ASTM certification requirements while not sacrificing aircraft engine performance in terms of the emissions and operability metrics key to Federal Aviation Administration (FAA) engine certification. Properties of the tuned fuels are measured/validated in the laboratory and communicated to the simulation team.

For these predictive simulations, the in-house computational fluid dynamics solver PeleLMeX is used. PeleLMeX is a proven exascale capable, CPU/GPU compatible, adaptive mesh refinement (AMR) reactive flow solver described in detail in the previous section. The PeleLMeX solver interacts with other libraries of the Pele suite of solvers such as PelePhysics (models different fluid equations of states, detailed transport

[b]https://www.energy.gov/eere/bioenergy/sustainable-aviation-fuel-grand-challenge

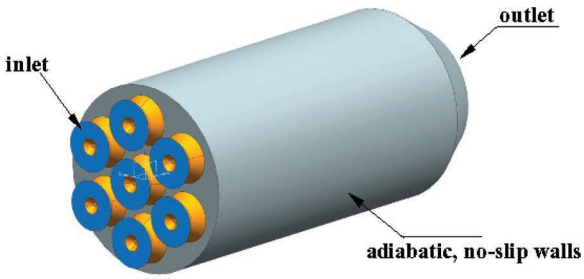

FIGURE 4. NASA seven-element, LDI test configuration and boundary conditions used in this study.

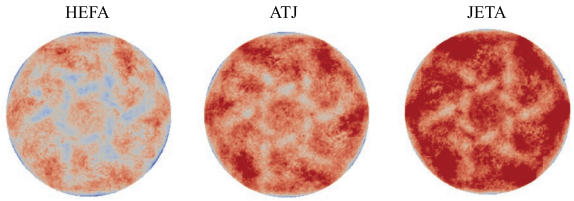

FIGURE 5. Mean droplet diameter (d_{10}) from three fuels observed at an axial plane inside the LDI combustor.

phenomena, liquid fuel sprays, soot formation and radiation heat transfer) and PeleAnalysis (handles post-processing of solution data).

As a part of this effort, the NREL team has been studying two combustor architectures, a seven-element lean direct injection (LDI) combustor configuration developed at NASA Glenn Research Center[9] and a proprietary combustor for commercial supersonic transport engines being developed by General Electric (GE) Aerospace and Georgia Institute of Technology through the FAA-funded ASCENT program.[c] The niche nature of commercial supersonic aviation provides a unique platform for the evaluation of fuel and combustion technologies that are currently radical for conventional subsonic commercial aircraft. This research collaboration with Georgia Institute of Technology leverages this opportunity to investigate the performance of 100% drop-in fuels, which, while not yet certified for commercial aviation, are anticipated to be critical for the operational requirements of supersonic combustion systems. Validation of the simulation results against experimental measurements is not addressed within the scope of this work. The reader is directed to forthcoming publications by this research group, which will present a comparison of the simulation results with experimental data obtained by the Georgia Institute of Technology.

The seven-element combustor test rig used for the first study is shown in Figure 4. Two SATFs, alcohol-to-jet (ATJ) and hydrotreated esters and fatty acids (HEFA), were chosen for the numerical study along with the conventional Jet-A fuel. Figure 5 shows the time-averaged droplet diameter of the SATFs on an axial plane located downstream of the fuel injection. The numerical simulations reveal that HEFA has smaller droplet sizes, leading to faster evaporation and lower residence times in comparison to ATJ and Jet-A fuels. This is attributed to the differences in thermo-physical properties of the fuels. The lower viscosity and density values of HEFA lead to a smaller droplet size distribution and hence a higher evaporation rate. Higher evaporation rate leads to a fuel rich mixture which impacts flame stabilization and most importantly emissions, NOx and soot, production. This study is a demonstration of how the impact of differences in thermophysical properties of the fuels on combustor performance can be studied effectively using predictive numerical simulations.

OXYCOMBUSTION FOR POWER GENERATION

Coupling electricity generation via combustion of hydrocarbons with CCUS is a promising pathway toward net-zero electricity generation. When combustion of biofuels is coupled to CCUS, carbon-negative electricity generation becomes possible. However, achieving these goals requires expensive reengineering of the power generation cycle. One viable approach is the Allam–Fetvedt cycle, as shown in Figure 6.[10] This is a closed-loop cycle using supercritical carbon dioxide (sCO_2) as a working fluid. Heat is provided by

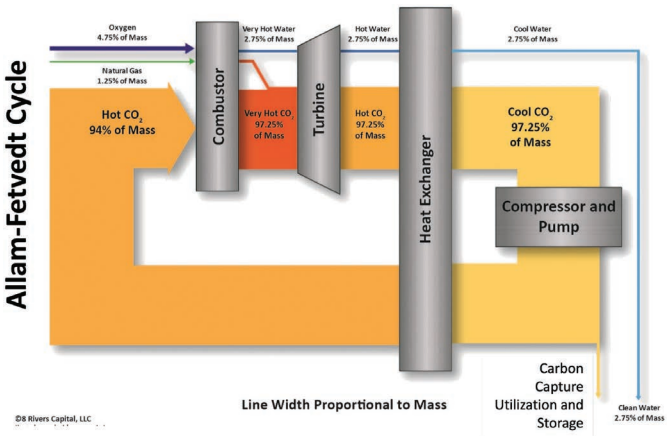

FIGURE 6. Allam–Fetvedt cycle.

[c]https://ascent.aero/project/low-emission-premixed-combustion-technology-for-supersonic-civil-transport/

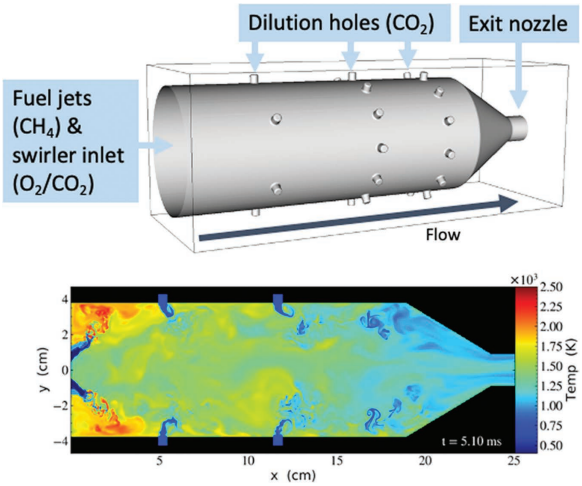

FIGURE 7. Combustor geometry and simulation of 40% CH4 jet with the Soave–Redlich–Kwong (SRK) equation of state.

combustion of hydrocarbons with oxygen at supercritical conditions around 300 bar, resulting in an exhaust stream of water and CO_2, allowing the easy removal of carbon dioxide for use or sequestration. Oxy-combustion completely sidesteps the expensive problem of reducing harmful NOx emissions. Additionally, increased density of CO_2 at supercritical conditions allows for a relatively compact design of the turbine used to produce power. Compared to existing combined cycle generation facilities, the new proposed cycle with sCO_2 is efficient and needs a smaller land footprint.

Reacting flow processes in traditional gas turbines with natural gas as fuel have been researched and optimized over several decades. The operating conditions for these gas turbines are in the range of 20–30 bar and with 10%–15% CO_2 concentration. However, the operating conditions for Allam–Fetvedt cycle combustors are in 300-bar range and with 90% CO_2 concentration. Knowledge about reacting flow phenomenon at these conditions are limited. Furthermore, experimental measurements to guide design studies are not seen in literature. For all these reasons, high-fidelity first-principles simulations are the only mechanism available to develop this technology.

Simulations of reacting flow in an sCO_2 environment requires the use of nonideal equations of state. Additionally, lack of a reduced-order model in the literature motivates the use of first-principles simulation with unprecedented spatial and temporal resolution. For all these reasons, simulation of these systems is truly an HPC problem needing exascale computing resources. Figure 7 shows simulations of the combustor for the Allam–Fetvedt cycle performed using PeleC using DOE HPC resources. This high-fidelity simulation was performed as a part of a High Performance Computing for Energy Innovation (HPC4EI) collaborative project between NREL and 8 Rivers, LLC.[11] The focus of these simulations was to develop knowledge and understanding of novel combustor-turbine technology for oxy-combustion with sCO_2. These HPC simulations are helping accelerate the development and deployment of this technology.

HYDROGEN FOR INDUSTRIAL DECARBONIZATION

Industrial processes account for approximately 30 % of U.S. air pollution emissions: an amount on the same order as electricity generation and transportation. Half of these emissions come from five industrial sectors: chemical manufacturing, petroleum refining, iron and steel, food and beverage, and cement. In many of these sectors, combustion processes are used to provide process heat at temperatures that cannot be achieved using electrical heating.[12] This makes reengineering these processes to use combustion of hydrogen and other low-carbon feedstocks a key research question.

An overview of the complex steelmaking process illustrates the challenges in introducing new technologies. Introducing newer fuels in direct iron ore reduction, and the adoption of electric arc furnaces, allow for significant reductions in pollutant emissions. There is a widespread use of combustion in steelmaking for preheating of scrap, raw materials, steelmaking vessels, and reheating of the steel for thermomechanical processing. Re-engineering of these processes is a significant technological challenge where simulations can play a critical role for making iron and steel industry efficient.

Substitution of hydrogen for natural gas or other hydrocarbons as a heat source is not a drop-in replacement process. Under fuel–lean conditions, employed to mitigate exhaust NOx emissions, the high mobility of the hydrogen fuel leads to flame instabilities. These instabilities can act synergistically with turbulence to accelerate the flame, potentially causing flashback, a critical safety and operability concern. This behavior depends strongly on local conditions and is therefore much more difficult to predict and control than flames burning fossil fuels. In addition, hydrogen's high mobility complicates its storage and transport for practical applications, typically requiring complete redesign of the combustor and fuel storage and delivery systems. Because of high development costs and safety concerns, simulation is expected to play a key role in enabling this transition.

CONCLUSION

Simulations of combustion using HPC and the enabled AI/ML approaches are playing a vital role in the development and deployment of new technologies essential for the energy transition in transportation, electricity generation, and industrial processes. The ability to perform these high-fidelity simulations is a direct result of significant investment in developing next-generation software tools such as the Pele codes through DOE's ECP. This software is currently being used to screen candidate SATFs and to enhance understanding of combustion in novel conditions that could enable affordable carbon capture. As new energy technologies are proposed, particularly in the area of industrial processes, the set of HPC tools described in this work will be extended and enhanced with capability to de-risk and hasten deployment of the new technologies. For example, the research teams at NREL are already creating exascale-ready software based on the existing tools to capture multiphase flow in steelmaking with direct reduction of iron ore.

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REFERENCES

1. M. J. Martin et al., "Integrating energy-efficient computing with computational research to accelerate energy technology," in *Comput. Sci. Eng.*, no. 1, pp. 1–10, PrePrints 5555, doi: 10.1109/MCSE.2025.3549359
2. M. T. Henry de Frahan et al., "The Pele simulation suite for reacting flows at exascale," *Proc. SIAM Conf. Parallel Process. Sci. Comput.*, pp. 13–25, 2024, doi: 10.1137/1.9781611977967.2.
3. M. T. Henry de Frahan et al., "PeleC: An adaptive mesh refinement solver for compressible reacting flows," *Int. J. High Perform. Comput. Appl.*, vol. 37, nos. 115–131, Sep. 2022, Art. no. 109434202211211.
4. L. Esclapez et al., "PeleLMeX: An AMR low Mach number reactive flow simulation code without level sub-cycling," *J. Open Source Softw.*, vol. 8, no. 90, 2023, Art. no. 5450, doi: 10.21105/joss.05450.
5. I. Barrio Sanchez, A. Almgren, J. Bell, M. Henry de Frahan, and W. Zhang, "A new re-redistribution scheme for weighted state redistribution with adaptive mesh refinement," *J. Comput. Phys.*, vol. 504, May 2024, Art. no. 112879, doi: 10.1016/j.jcp.2024.112879.
6. A. Giuliani, A. Almgren, J. Bell, M. Berger, M. Henry de Frahan, and D. Rangarajan, "A weighted state redistribution algorithm for embedded boundary grids," *J. Comput. Phys.*, vol. 464, Sep. 2022, Art. no. 111305, doi: 10.1016/j.jcp.2022.111305.
7. M. T. Henry de Frahan et al., "Deep reinforcement learning for dynamic control of fuel injection timing in multi-pulse compression ignition engines," *Int. J. Engine Res.*, vol. 23, no. 9, pp. 1503–1521, 2022, doi: 10.1177/14680874211019345.
8. N. T. Wimer, M. T. Henry de Frahan, and S. Yellapantula, "Deep reinforcement learning to discover multi-fuel injection strategies for compression ignition engines," *Int. J. Engine Res.*, vol. 24, no. 9, pp. 3985–4007, 2023, doi: 10.1177/14680874231184002.
9. K. M. Tacina, "Swirl-venturi lean direct injection combustion technology for low-nox aero gas turbine engines," presented at the Spring Tech. Meeting Central States Sect. Combustion Inst., Minneapolis, MN, USA, 2018, no. GRC-E-DAA-TN55621.
10. R. Allam et al., "Demonstration of the Allam cycle: An update on the development status of a high efficiency

supercritical carbon dioxide power process employing full carbon capture," *Energy Procedia*, vol. 114, pp. 5948–5966, Jul. 2017, doi: 10.1016/j.egypro.2017.03.1731.

11. M. T. Henry de Frahan et al., "Simulation of methane oxycombustion in supercritical carbon dioxide," in *Proc. Turbo Expo: Turbomachinery Tech. Conf. Expo.*, American Society of Mechanical Engineers, 2023, p. 1–8.
12. J. Cresko et al., "U.S. department of energy's industrial decarbonization roadmap," USDOE Office of Energy Efficiency and Renewable Energy (EERE), Washington, DC, USA, Sep. 2022. [Online]. Available: https://www.osti.gov/biblio/1961393

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